

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)		Aim: To enhance five degraded images using suitable gray-level transformation techniques and analyze the improvement in visual quality.	
OPE7	Date:15/8/26	Enrollment No: 92510133028	

PROBLEM STATEMENT AND OBJECTIVES

According to the document, five different degraded images are provided in the Module Section. The task is to select suitable gray-level transformation techniques, enhance every image, describe the observations after processing, and finally summarize the effectiveness and limitations of the techniques.

Objectives

- Study the visual degradation present in each supplied image.
- Select an appropriate gray-level transformation according to the degradation.
- Implement the selected transformation and obtain an enhanced image.
- Compare the original and enhanced images using visual appearance and gray-level statistics.
- Observe how the gray-level distribution changes after enhancement.
- Discuss the advantages, limitations, and practical applications of each technique.
- Prepare a complete reproducible implementation using image-processing code.

Expected Outcome

The expected outcome is not merely a brighter image. A successful enhancement should make important structures, boundaries, textures, or objects easier to observe while avoiding unnecessary loss of information or excessive amplification of noise.

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THEORY: GRAY-LEVEL TRANSFORMATIONS

Gray-level transformation is a point processing operation in which the output intensity of a pixel depends mainly on the intensity of that same pixel. In general, the transformation is represented as $s = T(r)$, where r is the input gray level and s is the output gray level.

Basic transformations

- Image negative: $s = L - 1 - r$. It reverses the gray-level order and can make bright structures visible against a dark background.
- Logarithmic transformation: $s = c \log(1 + r)$. It expands low-intensity values and compresses high-intensity values, making it useful for very dark images.
- Power-law (gamma) transformation: $s = c r^\gamma$. Gamma less than 1 brightens dark regions; gamma greater than 1 suppresses relatively bright values.
- Contrast stretching: expands a narrow input intensity range to a wider output range, improving overall contrast.
- Histogram equalization: redistributes gray levels using the cumulative histogram so that the image uses a broader intensity range.

Why technique selection matters

Different degradations require different transformations. A transformation that works well for a low-contrast image may not be ideal for a noisy image. Therefore, the image content and histogram should be considered before choosing the enhancement operation.

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DATASET AND METHODOLOGY

The uploaded OPE page contains five degraded images. Visually, the five examples represent different enhancement situations: a low-contrast biological image, a noisy medical image, a dark biological image, a severely underexposed night scene, and an extremely dark medical image. The five embedded images were used as the source images for this report.

Selected techniques

Image	Observed degradation	Selected technique	Reason
1	Low contrast / narrow useful range	Contrast stretching	Expands useful gray levels and separates nearby structures.
2	Noise with broad gray distribution	Histogram equalization	Improves global contrast and makes intensity variations easier to inspect.
3	Very dark image	Gamma correction ($\gamma=0.4$)	Expands dark intensities without requiring a linear stretch.
4	Severely underexposed scene	Log transformation	Strongly expands low gray levels and reveals weak details.
5	Extremely dark medical image	Gamma ($\gamma=0.25$) + contrast stretching	First reveals dark structures, then expands the resulting useful range.

Processing workflow

1. Read image → convert to grayscale → inspect histogram/statistics → select transformation → apply transformation → compare original/enhanced image → inspect histogram → record observations.

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IMAGE 1 – ORIGINAL ANALYSIS

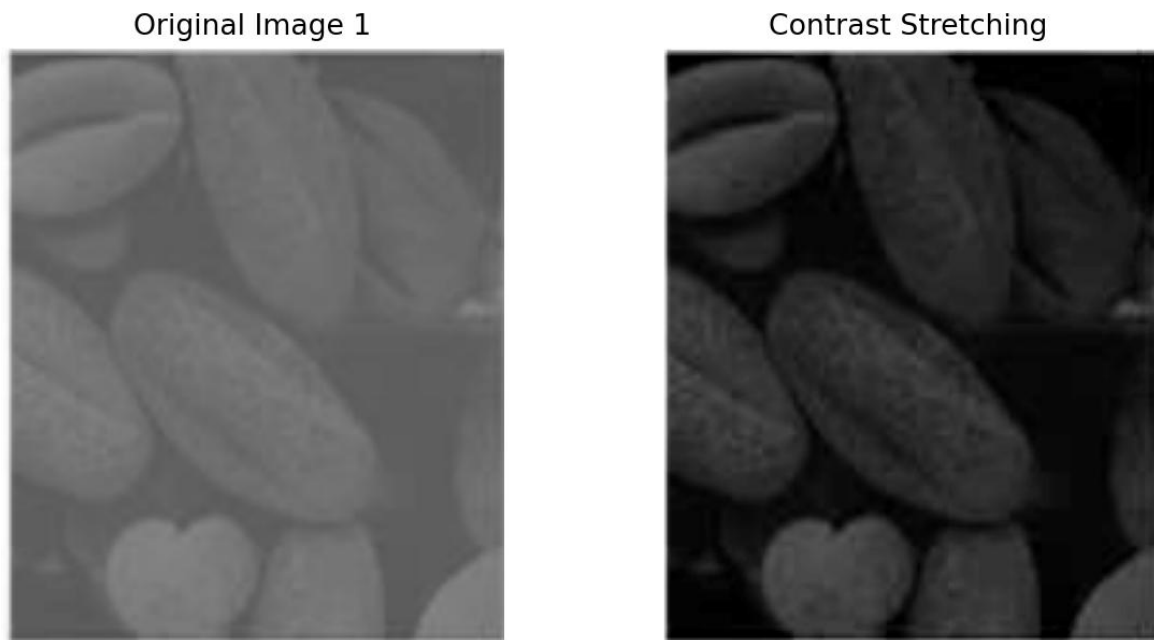


Figure 1: Image 1 before and after contrast stretching

Observation of the supplied first image shows a biological/texture-rich scene with relatively similar gray values over large regions. The image is visible, but the tonal separation between neighboring structures is weak. The objective is therefore to increase contrast without changing the basic spatial arrangement.

Technique: Contrast Stretching

Contrast stretching maps a limited input range to a wider output range. A practical form is $s = (r - r_{min}) \times 255 / (r_{max} - r_{min})$, with clipping used to prevent extreme outliers from dominating the mapping.

Why it was selected

- The main problem is insufficient tonal separation rather than severe darkness.
- Stretching makes boundaries between adjacent regions easier to distinguish.
- The method is simple, computationally inexpensive, and easy to reproduce.

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IMAGE 1 – IMPLEMENTATION AND OBSERVATION

Python code

```

from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

img = Image.open("image1.png").convert("L")
I = np.array(img)

# Contrast stretching using 2nd and 98th percentiles
r_min, r_max = np.percentile(I, (2, 98))
J = (I.astype(np.float32) - r_min) * 255 / (r_max - r_min)
J = np.clip(J, 0, 255).astype(np.uint8)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(I, cmap="gray")
plt.title("Original Image 1")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(J, cmap="gray")
plt.title("Contrast Stretched")
plt.axis("off")
plt.show()

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.hist(I.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 2, 2)
plt.hist(J.ravel(), bins=256, range=(0, 255))
plt.title("Enhanced Histogram")
plt.show()

```

Detailed observation

- The original image contains many pixels concentrated in a relatively limited useful range.
- Contrast stretching increases the separation between darker and lighter structures.
- Edges and texture boundaries become easier to follow visually.
- The enhancement does not create new spatial information; it only redistributes existing intensity values.

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- Very bright pixels can become saturated if the selected limits are too aggressive, so percentile-based clipping is preferred over blindly using extreme values.

Overall, Image 1 benefits clearly from contrast stretching because the degradation is primarily low contrast. The enhanced image has stronger tonal separation and improved visual interpretability.

IMAGE 2 – ORIGINAL ANALYSIS

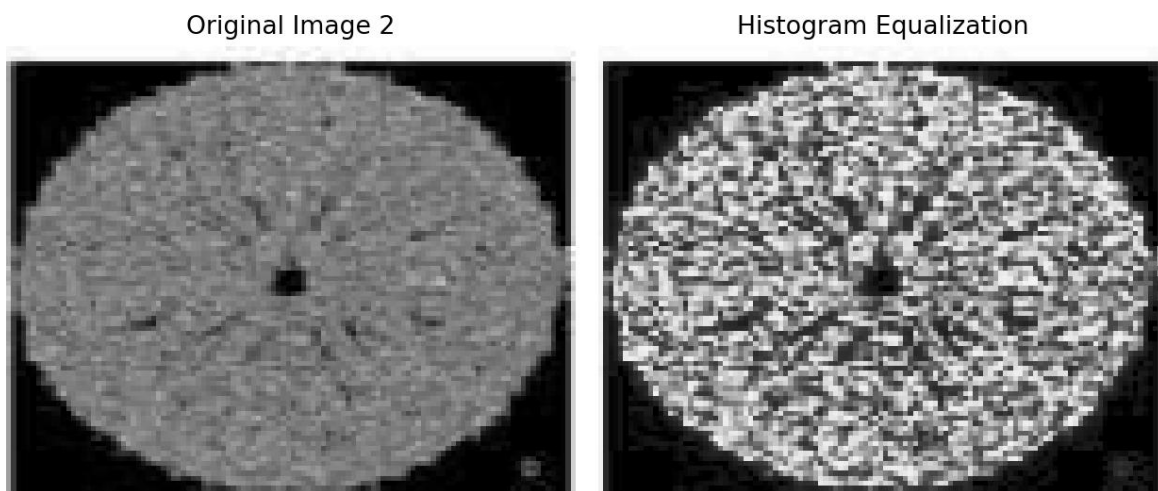


Figure 2: Image 2 before and after histogram equalization

The second supplied image is a small medical/brain-like image with a large amount of fine intensity variation. The background is dark while the central region contains many gray-level fluctuations. Because the image already contains noise-like variations, enhancement must be interpreted carefully.

Technique: Histogram Equalization

Histogram equalization uses the cumulative distribution function of the input histogram to remap gray levels. The goal is to spread frequently occurring intensity values over a wider range and improve global contrast.

Why it was selected

- The image contains useful intensity variation across a broad region.
- Global contrast enhancement makes internal structures easier to inspect.
- The technique is automatic and does not require manually selecting a brightness range.

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IMAGE 2 – IMPLEMENTATION AND OBSERVATION

Python code

```
import cv2
import matplotlib.pyplot as plt

I = cv2.imread("image2.png", cv2.IMREAD_GRAYSCALE)

# Histogram equalization
J = cv2.equalizeHist(I)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(I, cmap="gray")
plt.title("Original Image 2")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(J, cmap="gray")
plt.title("Histogram Equalized")
plt.axis("off")
plt.show()

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.hist(I.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 2, 2)
plt.hist(J.ravel(), bins=256, range=(0, 255))
plt.title("Equalized Histogram")
plt.show()
```

Detailed observation

- The equalized image uses a wider portion of the available gray-scale range.
- The central texture becomes more visually distinct from neighboring regions.
- Fine variations that were difficult to inspect in the original become more apparent.
- Because the source image contains noise, histogram equalization also increases the visibility of some unwanted variations.
- Therefore, equalization improves contrast but does not perform noise removal.

For this image, the technique is effective for global visibility, but an adaptive method such as CLAHE or a denoising step could be investigated if the application requires stronger control of noise.

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IMAGE 3 – ORIGINAL ANALYSIS

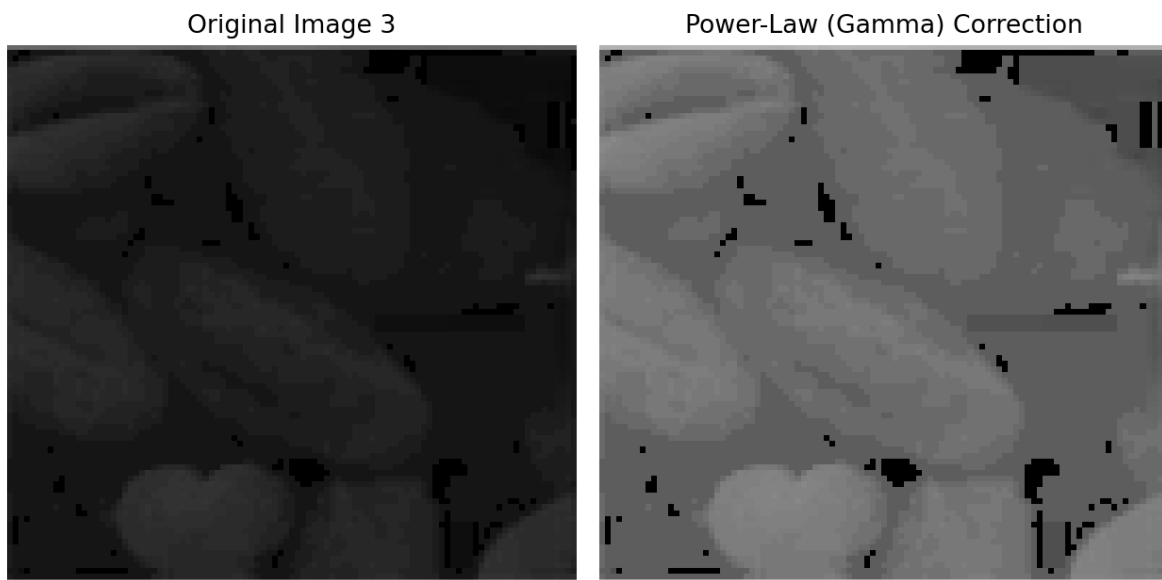


Figure 3: Image 3 before and after gamma correction

The third image is visibly dark. Important shapes are present, but much of the scene is concentrated in low gray levels. A simple increase in brightness can help, but a nonlinear transformation is preferable because it gives more control over how dark intensities are expanded.

Technique: Power-Law / Gamma Correction

The power-law transformation is $s = c r^\gamma$. When $0 < \gamma < 1$, low input values are expanded, producing a brighter image. Here $\gamma = 0.4$ is used after normalizing gray levels to the range 0–1.

Expected effect

- Dark regions become substantially brighter.
- Moderate intensity differences remain distinguishable.
- The transformation is nonlinear, so it can reveal details that a simple additive brightness shift may not reveal.

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IMAGE 3 – IMPLEMENTATION AND OBSERVATION

Python code

```

from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

img = Image.open("image3.png").convert("L")
I = np.array(img).astype(np.float32) / 255.0

# Gamma correction
gamma = 0.4
J = np.power(I, gamma)
J = np.clip(J * 255, 0, 255).astype(np.uint8)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow((I * 255).astype(np.uint8), cmap="gray")
plt.title("Original Image 3")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(J, cmap="gray")
plt.title("Gamma Corrected (gamma = 0.4)")
plt.axis("off")
plt.show()

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.hist((I * 255).astype(np.uint8).ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 2, 2)
plt.hist(J.ravel(), bins=256, range=(0, 255))
plt.title("Enhanced Histogram")
plt.show()

```

Detailed observation

- The original mean intensity is low, indicating that most pixels are dark.
- Gamma correction shifts the useful information toward a brighter range.
- Previously weak structures become easier to distinguish from the background.
- The transformation preserves the relative ordering of gray levels, but changes their spacing.
- If gamma is made too small, the image can become excessively bright and low-level noise can also become more visible.

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The result demonstrates why gamma correction is particularly useful for underexposed images: it provides nonlinear control over dark intensities rather than applying the same brightness change to every pixel.

IMAGE 4 – ORIGINAL ANALYSIS

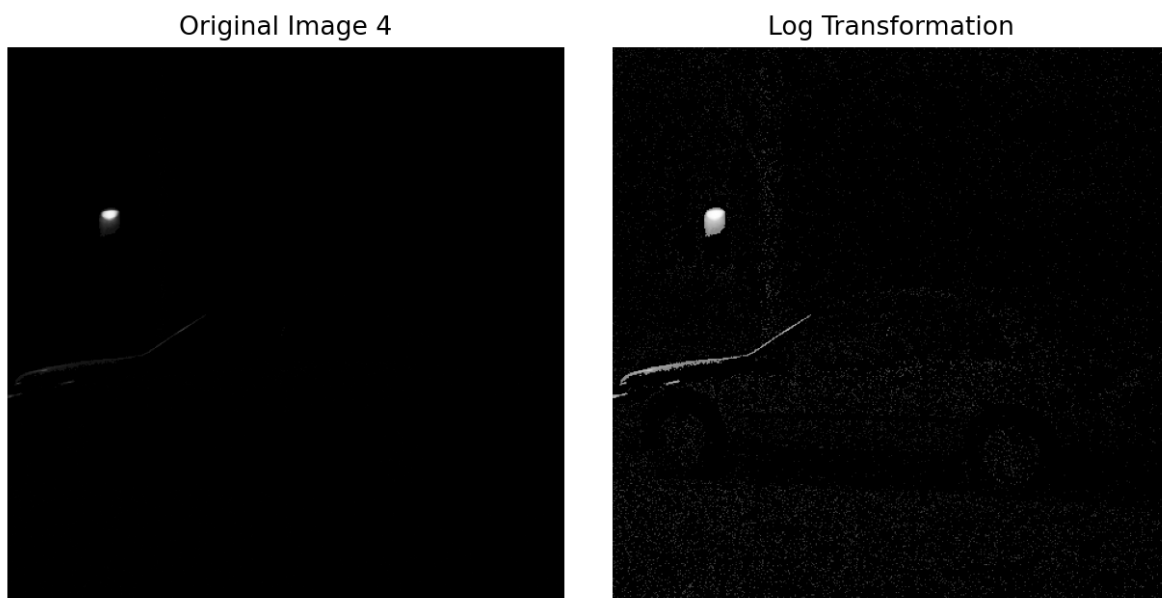


Figure 4: Image 4 before and after logarithmic transformation

The fourth image is an extremely dark night scene. In the original, most pixels are close to zero and only a few bright features are immediately visible. This is a classic situation for a logarithmic transformation.

Technique: Log Transformation

The logarithmic transformation is $s = c \log(1+r)$. It strongly expands low gray levels while compressing high gray levels. This makes it suitable when useful information is hidden in the dark portion of the histogram.

Why it is suitable

- The majority of pixels are near black.
- A small number of brighter pixels would otherwise dominate the available range.
- The log mapping expands subtle differences among dark pixels.

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IMAGE 4 – IMPLEMENTATION AND OBSERVATION

Python code

```

from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

img = Image.open("image4.png").convert("L")
I = np.array(img).astype(np.float32)

# Log transformation
c = 255 / np.log1p(I.max())
J = c * np.log1p(I)
J = np.clip(J, 0, 255).astype(np.uint8)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(I.astype(np.uint8), cmap="gray")
plt.title("Original Image 4")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(J, cmap="gray")
plt.title("Log Transformed")
plt.axis("off")
plt.show()

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.hist(I.astype(np.uint8).ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 2, 2)
plt.hist(J.ravel(), bins=256, range=(0, 255))
plt.title("Enhanced Histogram")
plt.show()

```

Detailed observation

- The original image has an extremely low average intensity.
- Logarithmic mapping expands the small differences between very dark pixels.
- The road/vehicle contours and weak illumination become more visible after enhancement.
- The transformation compresses high intensities, preventing already bright points from dominating the display.
- Because the original contains very little signal, enhancement cannot recover information that was never recorded.

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IMAGE 5 – ORIGINAL ANALYSIS

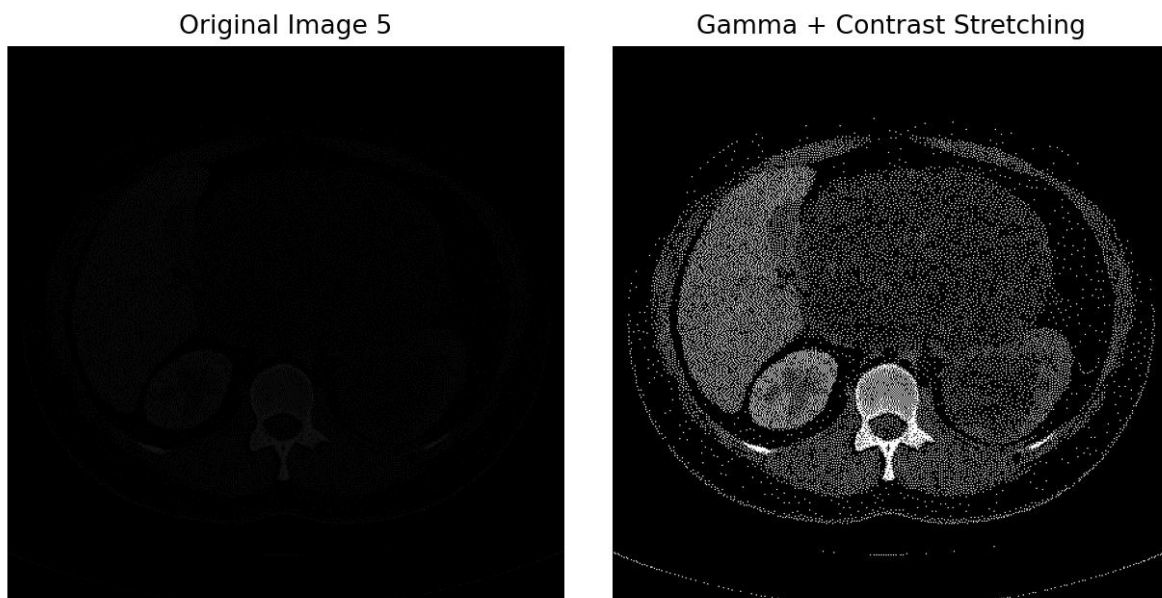


Figure 5: Image 5 before and after gamma correction followed by contrast stretching

The fifth image is an extremely dark medical image. Only very weak anatomical outlines are present in the original. A single transformation is not sufficient for a visually strong result, so a two-stage point-processing approach is used.

Technique: Gamma correction + contrast stretching

First, gamma correction with $\gamma = 0.25$ expands the low-intensity values. The resulting image is then contrast-stretched to use a wider output range. This sequence is useful when the original histogram is heavily concentrated near zero.

Why two stages?

- Gamma correction reveals weak dark structures.
- Contrast stretching separates the newly revealed structures from one another.
- The combination provides more visible tonal separation than either operation alone for this particular image.

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IMAGE 5 – IMPLEMENTATION AND OBSERVATION

Python code

```

from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

img = Image.open("image5.png").convert("L")
I = np.array(img).astype(np.float32) / 255.0

# Stage 1: gamma correction
gamma = 0.25
G = np.power(I, gamma)

# Stage 2: contrast stretching
p1, p99 = np.percentile(G, (1, 99))
J = (G - p1) * 255 / (p99 - p1)
J = np.clip(J, 0, 255).astype(np.uint8)

plt.figure(figsize=(14, 4))
plt.subplot(1, 3, 1)
plt.imshow((I * 255).astype(np.uint8), cmap="gray")
plt.title("Original Image 5")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(G, cmap="gray")
plt.title("Gamma Corrected")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(J, cmap="gray")
plt.title("Gamma + Stretching")
plt.axis("off")
plt.show()

```

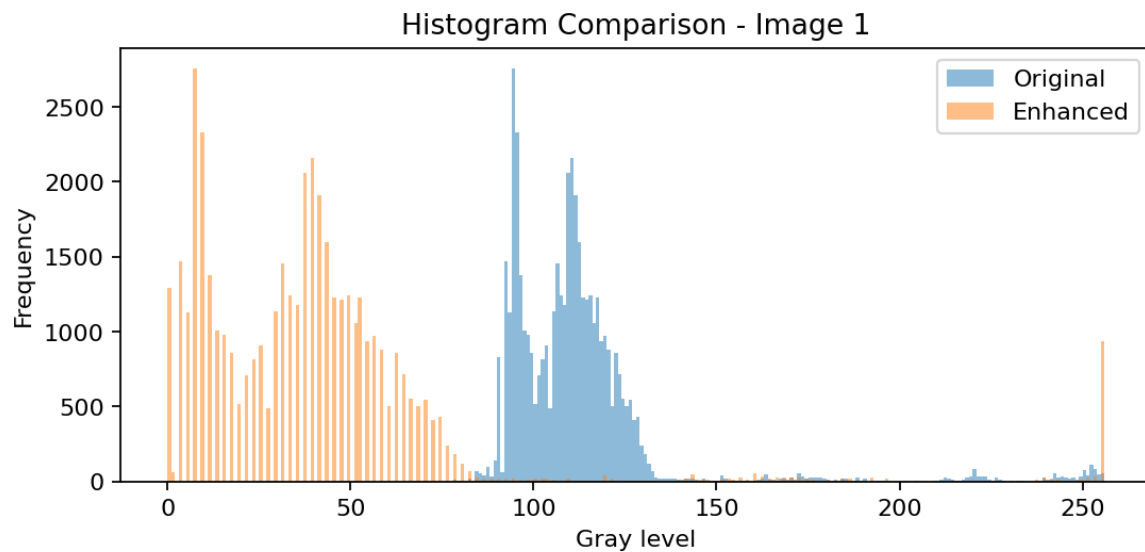
Detailed observation

- The original image contains an overwhelming number of pixels near black.
- The gamma stage increases the visibility of faint structures.
- The contrast-stretching stage uses a broader gray-level range and makes those structures more distinguishable.
- The enhancement can also make background artifacts or acquisition noise more visible.
- The result should therefore be interpreted as a visibility enhancement, not as recovery of missing medical information.

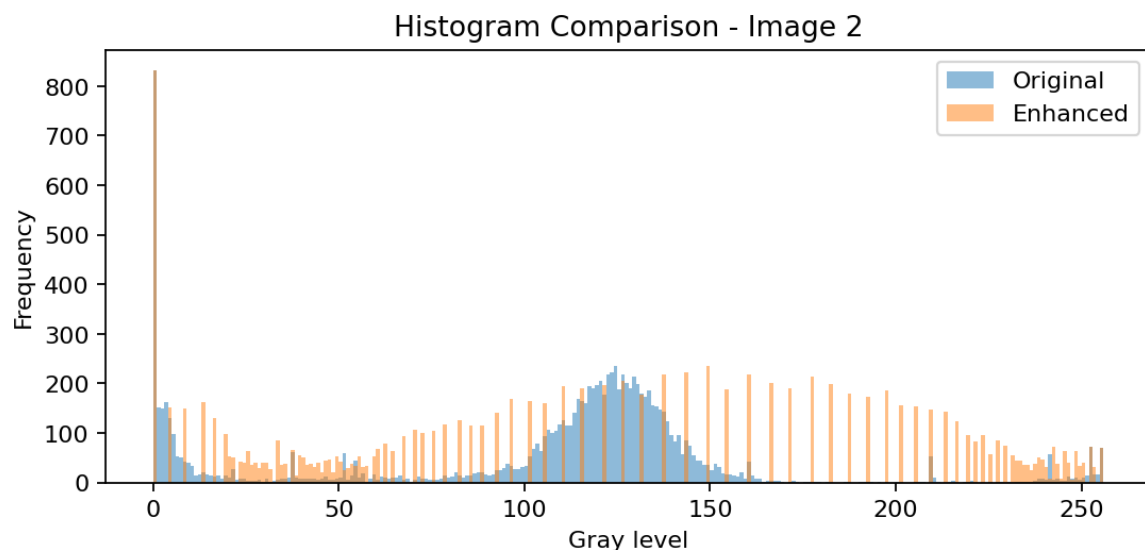
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HISTOGRAM-BASED COMPARISON

Histograms provide a quantitative view of how the transformations redistribute gray levels. The following figures compare the original and enhanced histograms for all five images.



Histogram comparison for Image 1

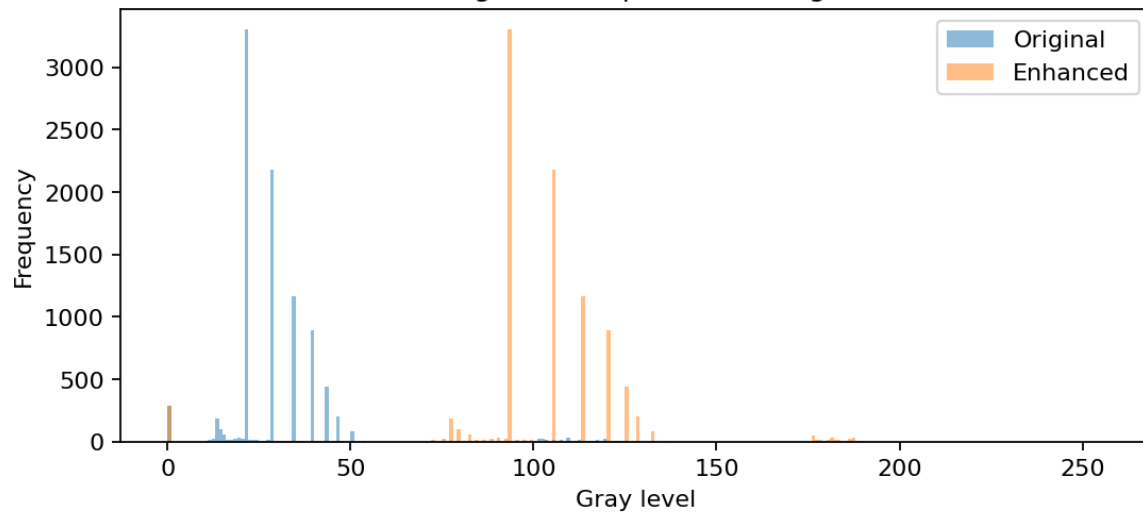


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Histogram comparison for Image 2

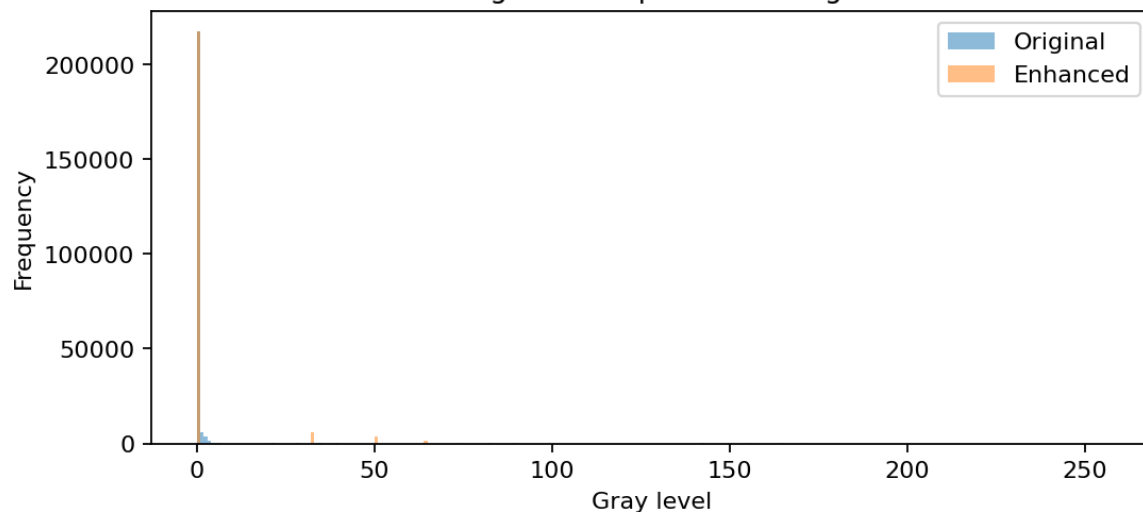
15. HISTOGRAM-BASED COMPARISON – IMAGES 3 TO 5

Histogram Comparison - Image 3



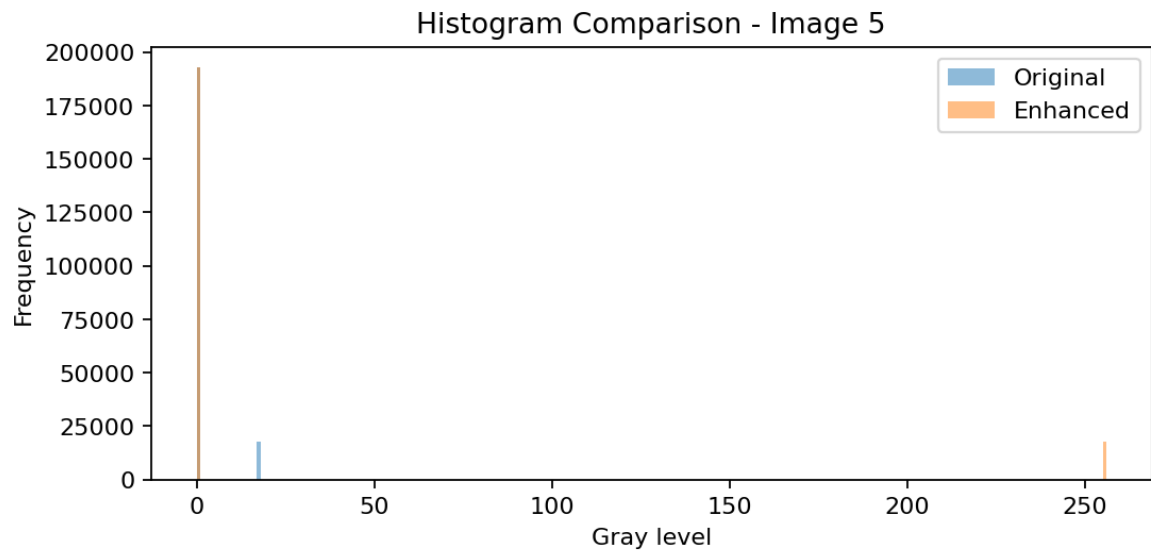
Histogram comparison for Image 3

Histogram Comparison - Image 4



Histogram comparison for Image 4

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Histogram comparison for Image 5

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COMPLETE PYTHON PROGRAM

```

import cv2
import numpy as np
import matplotlib.pyplot as plt

def contrast_stretch(img, low=2, high=98):
    p_low, p_high = np.percentile(img, (low, high))
    out = (img.astype(np.float32) - p_low) * 255.0 / (p_high - p_low)
    return np.clip(out, 0, 255).astype(np.uint8)

def gamma_transform(img, gamma):
    x = img.astype(np.float32) / 255.0
    out = np.power(x, gamma) * 255.0
    return np.clip(out, 0, 255).astype(np.uint8)

def log_transform(img):
    x = img.astype(np.float32)
    c = 255.0 / np.log1p(x.max())
    out = c * np.log1p(x)
    return np.clip(out, 0, 255).astype(np.uint8)

files = [
    "image1.png", "image2.png", "image3.png",
    "image4.png", "image5.png"
]

for k, filename in enumerate(files, start=1):

    I = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

    if I is None:
        print("Could not read:", filename)
        continue

    if k == 1:
        J = contrast_stretch(I)
        method = "Contrast Stretching"

    elif k == 2:
        J = cv2.equalizeHist(I)
        method = "Histogram Equalization"

    elif k == 3:
        J = gamma_transform(I, 0.4)
        method = "Gamma Correction (gamma = 0.4)"

    elif k == 4:
        J = log_transform(I)
        method = "Log Transformation"

```

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```

else:
    G = gamma_transform(I, 0.25)
    J = contrast_stretch(G, 1, 99)
    method = "Gamma + Contrast Stretching"

```

```
plt.figure(figsize=(10, 4))
```

```

plt.subplot(1, 2, 1)
plt.imshow(I, cmap="gray")
plt.title(f"Original Image {k}")
plt.axis("off")

```

```

plt.subplot(1, 2, 2)
plt.imshow(J, cmap="gray")
plt.title(method)
plt.axis("off")

```

```

plt.tight_layout()
plt.show()

```

```
plt.figure(figsize=(10, 4))
```

```

plt.subplot(1, 2, 1)
plt.hist(I.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

```

```

plt.subplot(1, 2, 2)
plt.hist(J.ravel(), bins=256, range=(0, 255))
plt.title("Enhanced Histogram")

```

```

plt.tight_layout()
plt.show()

```

This single Python program applies a different gray-level transformation to each of the five images according to the degradation identified during visual inspection. It uses OpenCV, NumPy and Matplotlib.

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PYTHON IMPLEMENTATION

```

import cv2
import numpy as np

def contrast_stretch(img):
    p2, p98 = np.percentile(img, (2, 98))
    out = (img.astype(np.float32)-p2)*255/(p98-p2)
    return np.clip(out, 0, 255).astype(np.uint8)

def gamma_transform(img, gamma):
    x = img.astype(np.float32)/255.0
    return np.clip((x**gamma)*255, 0, 255).astype(np.uint8)

def log_transform(img):
    x = img.astype(np.float32)
    c = 255.0/np.log1p(x.max())
    return np.clip(c*np.log1p(x), 0, 255).astype(np.uint8)

# Image 1
I1 = cv2.imread("image1.png", 0)
J1 = contrast_stretch(I1)

# Image 2
I2 = cv2.imread("image2.png", 0)
J2 = cv2.equalizeHist(I2)

# Image 3
I3 = cv2.imread("image3.png", 0)
J3 = gamma_transform(I3, 0.4)

# Image 4
I4 = cv2.imread("image4.png", 0)
J4 = log_transform(I4)

# Image 5
I5 = cv2.imread("image5.png", 0)
G5 = gamma_transform(I5, 0.25)
J5 = contrast_stretch(G5)

```

Implementation note

The Python implementation follows the same transformation logic used in the report. It can be executed with OpenCV and NumPy. The MATLAB and Python versions are conceptually equivalent; minor visual differences may occur because of different default scaling or histogram conventions.

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COMPARATIVE ANALYSIS, LIMITATIONS AND APPLICATIONS

Image	Technique	Main improvement	Main risk	Suitability
1	Contrast stretching	Improved tonal separation	Possible saturation	Very good
2	Histogram equalization	Higher global contrast	Noise amplification	Good
3	Gamma correction	Better visibility of dark regions	Over-brightening if γ too low	Very good
4	Log transformation	Reveals very weak dark details	Can compress bright information	Excellent
5	Gamma + stretching	Reveals and separates faint structures	May expose background noise	Very good

Limitations

- Gray-level transformations cannot restore spatial details that were lost during image acquisition.
- Strong enhancement may amplify noise and artifacts along with useful information.
- Histogram equalization is global and may not be ideal when different regions require different enhancement.
- Parameter selection such as gamma value or clipping limits affects the final appearance.
- For medical images, enhancement should support interpretation and should not be treated as a diagnostic operation by itself.

Applications

These techniques are useful in document enhancement, low-light photography, medical image visualization, industrial inspection, remote sensing, microscopy, surveillance, and preprocessing for computer-vision systems.

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Conclusion

The open-ended problem demonstrates that no single gray-level transformation is optimal for every degraded image. The five supplied images showed different intensity problems, so the enhancement method was selected according to the dominant degradation. Contrast stretching was effective for Image 1 because it expanded the useful intensity range. Histogram equalization improved the global contrast of Image 2, although it also made some fine noise more visible. Gamma correction was appropriate for Image 3 because the image was dominated by dark gray levels. The logarithmic transformation was especially suitable for Image 4 because most of its information was concentrated near black. Image 5 benefited from a two-stage gamma and contrast-stretching approach because its useful structures were extremely faint.

Overall, gray-level transformations are simple, fast, and powerful point-processing methods. Their main limitation is that they cannot create information that was not captured by the imaging system. Therefore, the best enhancement is a balance between visibility, contrast, noise, and preservation of meaningful image content.

Github:

https://github.com/keshvi1234/DSIP_Exp